

Collaborative Data Exploration Through Visualization: Analyzing Impact Of Conversational Assistants

Thanks to

**Abari
Bhattacharya**

(her thesis – just
defended on Apr 29,
2020)



Exploratory Data Analysis^[1]



Method for analyzing data predominantly using graphical techniques



Gain valuable insights into the information that can be gathered from the data

[1] John W. Tukey. Exploratory Data Analysis. Vol. 2. Reading, Mass: Addison-Wesley, 1977.

Exploratory Data Analysis is a Collaborative Process

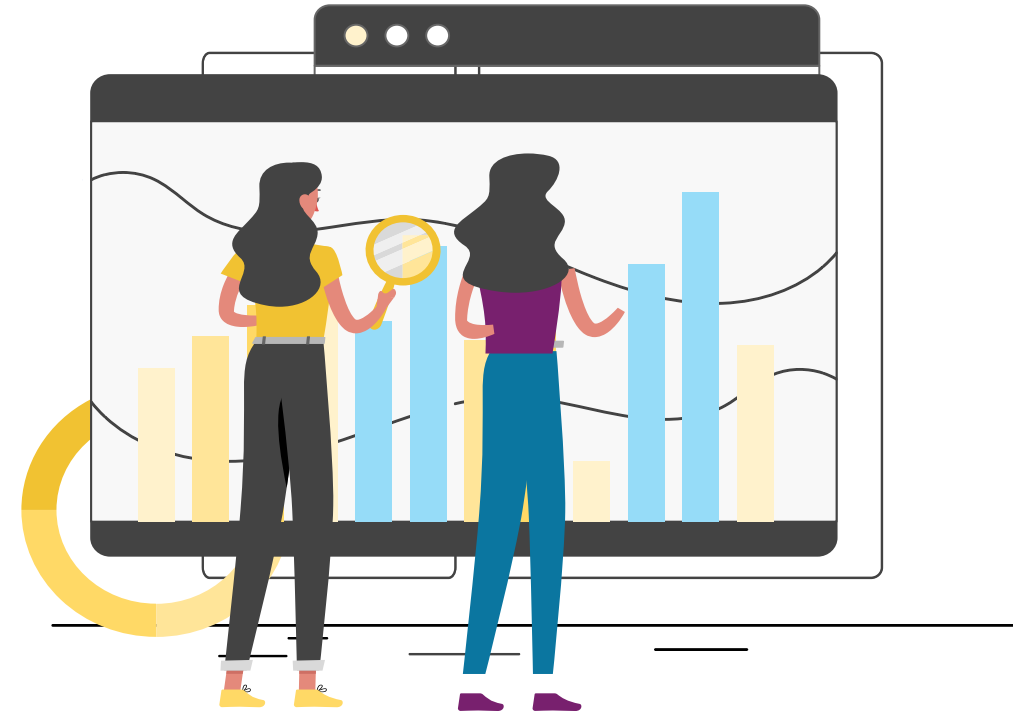
Multiple people work together to interpret data

Leverage the diverse perspectives

Enhance sense-making and discover insights

Traditional tools (Tableau , MS Excel , Plotly etc.)
→ designed typically for single user

Challenge for collaborative exploratory analysis of
data through visualizations^[3,4,5]



[3]Petra Isenberg et al. "Collaborative visualization: Definition, challenges, and research agenda". In: Information

[4]Wesley Willett et al. "CommentSpace: structured support for collaborative visual analysis". CHI '11.

[5]Do-Hyung Jeong, Soo-Young Ji, Evan A. Suma, et al. "Designing a collaborative visual analytics system to support users' continuous analytical processes. Human-centric Computing and Information.

Conversational Assistant for Data Visualization

Timeline:

Early Research

- Cox et al. (2001)
- Articulate, Sun et al (2010)

NLP Milestones

- Word Embeddings (2013)
- Attention (2017)
- BERT(2018)

LLM-based NLIs

- JarviX(2023)
- Chat2Vis (2023)
- LIDA (2023)
- VIST5 (2023)
- ChartAssistant (2024)
- WaitGPT (2024)
- Data Formulator 2 (2025)

Commercial Tools

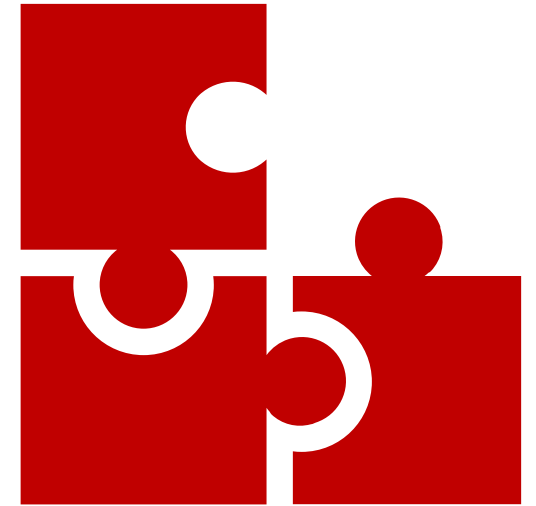
- Apple Siri(2011)
- IBM Watson Analytics

NLIs for Visualization

- Evizon(2018)
- Flowsense (2019)
- Articulate 2

Research Gap

- ✗ Not Interactive or Conversational (Chat2Vis, JarviX)
- ✗ Not Trained with Conversational Data (ncNet, ChartAssistant)
- ✗ Use of Complex Prompts(Chat2Vis)
- ✗ Not Evaluated with User Studies(Chat2Vis, LIDA, ChartAssistant)
- ✗ Limited or no focus on collaborative visualization or visual analytics(Chat2Vis, LIDA, JarviX, ncNet, VIST5,

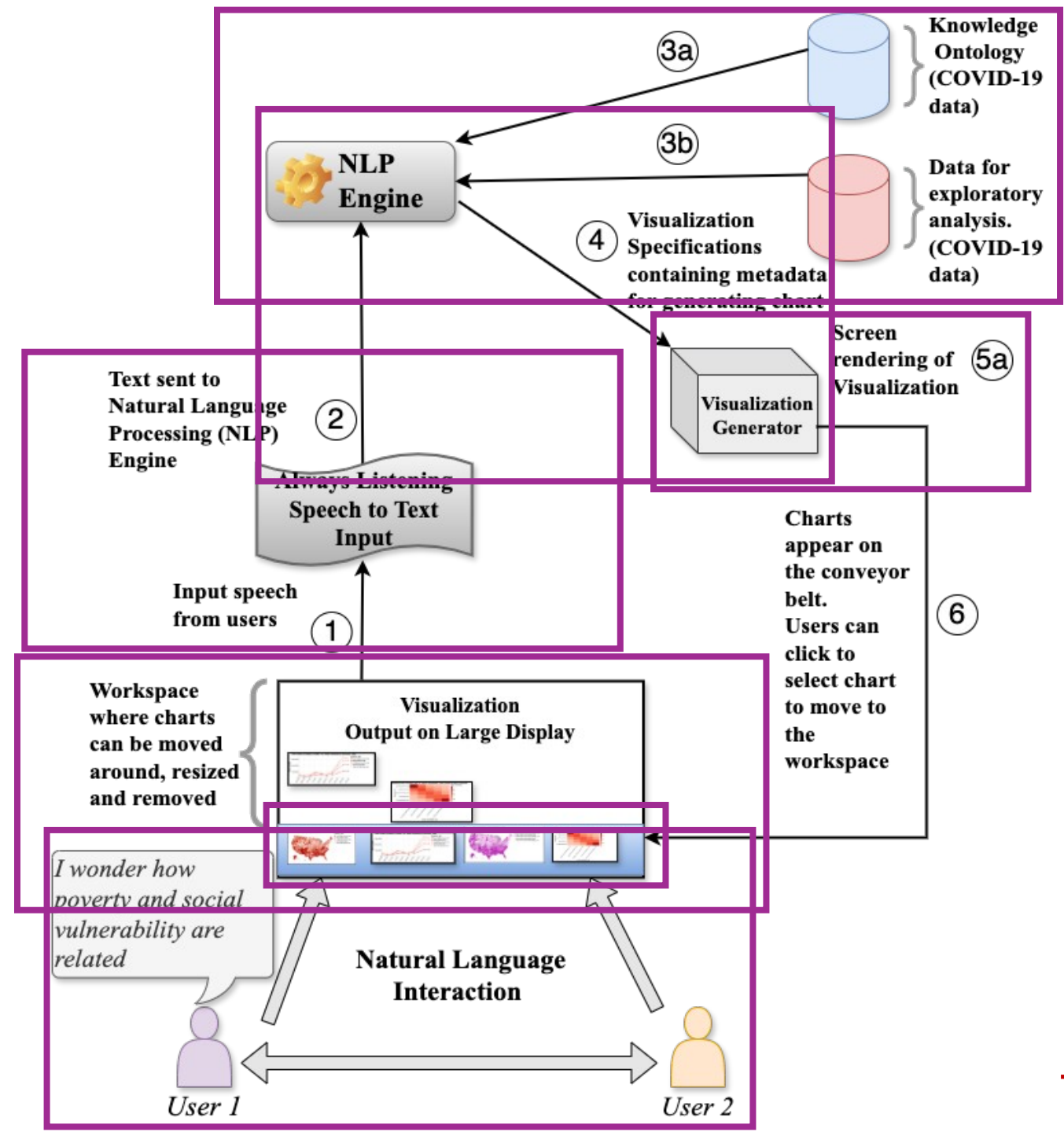




Part - 1

Investigating the impacts of a CA on users in a collaborative EDA scenario.

A Conversational Assistant for COVID-19 Data Exploration^[9]



User Study Setup

15 Groups of 2 participants

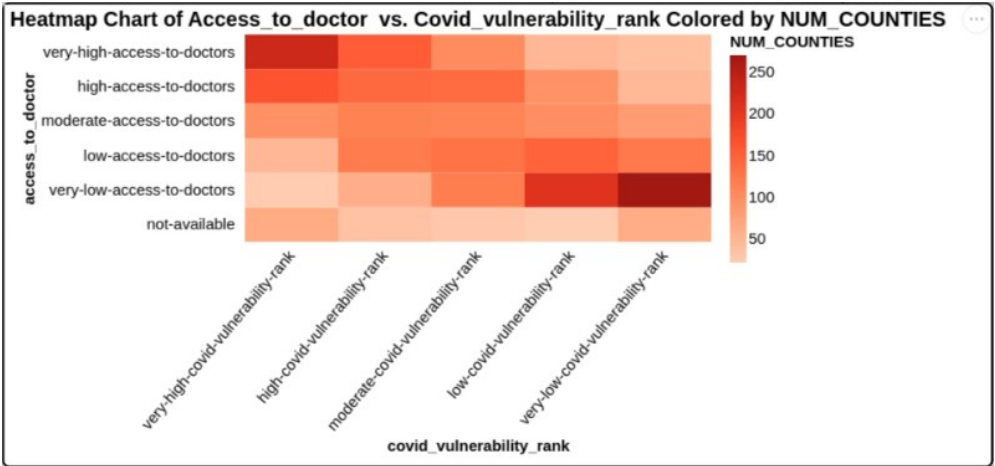
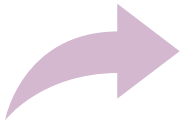
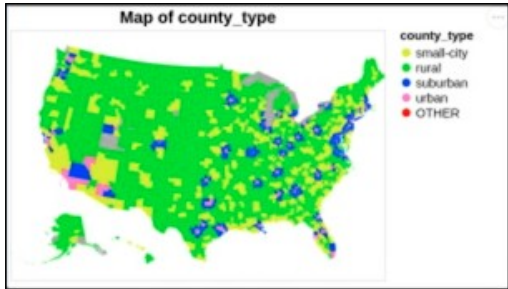
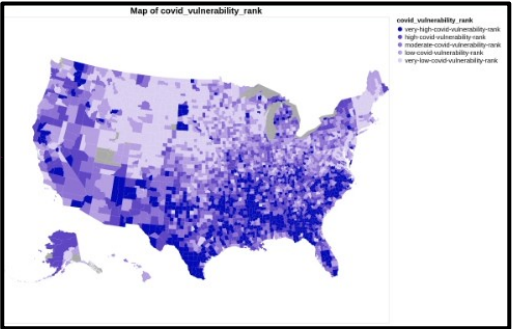
Explored COVID-19 Data For U.S. Counties

Timed Tasks 1 (25 minutes)	What regions and area types in the US should we prioritize for directing resources based on COVID risk, Poverty, Diabetes, Cardiovascular disease, Elderly population, Uninsured, Access to doctors?
Timed Task 2 (25 minutes)	Are there interesting patterns within different sets of counties that might help us understand COVID-19 outbreaks and better prioritize resources?
• Open-ended tasks • Audio and video recording • Users not given instructions on using interface • Encouraged to discuss the exploration • At the end of each task, participants shared their conclusions as a group • Post-study semi-structured interview	
Comparative Analysis with a Parallel Study	Supplementing our Analysis Our CA → Dialogue Manager (DM) Parallel Study → Property Extractor (PE)



A snippet of the interaction...

	Conversation	Chart Generated
1	U1: I think we can do Covid risk first, right?	Choropleth Map by Covid Vulnerability Rank
2	U2: We have a very high Covid rank.	-
3	U2: What do you think about it?	-
4	U1: Should we get a map of the county type first?	Choropleth Map by County Type
5	U1: Actually, it doesn't say that, well, what is the ongoing problem right now in America?	-
6	U2: The first is poverty, right?	-
7	U2: So, I think, wait.	-
8	U1: Well, I think we can do like Covid risk and then access to doctors and then...	Heat Map by Access to Doctor, Covid



Research Questions (RQ): how CA impacts users in a collaborative EDA scenario. Two CAs: Dialogue Manager (this one) vs simpler Property Extractor

RQ1 (Functionality) How does the functionality of the CA influence user behavior?	RQ 2 (Interpretability) How are the inferences drawn by users from their explorations affected by the CA?	RQ3 (Usability) How are users' perceptions of the usability of the NLI impacted by the CA?
- Different NLP engines produced different interaction patterns - More proactive system (PE) → guide exploration toward specific properties - Slower system (DM) → more time to reflect and explore more broadly	- Users reached different conclusions under the two systems - System behavior influenced what users focused on during exploration - CA design can shape users' exploratory paths and inferences	- Generated charts were generally seen as useful (72%) - Ease of use was mixed(40% Easy/Very Easy, 22% Neutral) - On-the-fly learning added cognitive load in collaborative tasks

Study 1 showed that CA design affected user behavior, user inferences, and perceived usability in collaborative EDA.

RQ1: How does **functionality** of the CA influence user behavior?

Two systems: PE (Property Extractor) / DM (Dialogue Manager)

Variables of interest

- Utterances processed
- Charts generated and selected
- Types of charts generated and selected
- Database Properties explored

A statistically significant difference between PE and DM in:

- **#Utterances processed**
- **#Charts generated**
- **#Line charts generated**

PE positively associates COVID-19 cases (Tasks 1 & 2)
-proactively generates charts
Line charts of COVID-19 trends by date.

DM shows more significant positive associations with database properties in Task 2.

DM's slower pace allows for user reflection.

PE's spontaneous chart generation drives focus on specific properties, potentially narrowing user exploration.

RQ2 (Interpretability): How are inferences drawn by users from their explorations affected by CA?

Variable of Interest

- Semantic similarity between conclusions drawn in PE and DM
 - Quantifies the similarity of meaning
 - Low similarity implies potential meaning divergence

Semantic overlap in conclusions reveals differences in task takeaways and explored “topics” between PE and DM.

PE: Participants mainly focus on exploring COVID-19 case counts across U.S. counties.

DM: Participants delve into other factors like diabetes risk and cardiovascular diseases.

Distinct conclusions drawn by PE and DM users confirm our hypothesis: system architecture influences the nature of conclusions in open-ended exploratory tasks.

Implication: Systems with tendencies toward specific visualizations shape users' exploration paths and conclusions.

RQ3: How are users' perceptions of **usability** of NLI impacted by CA?

Variables of Interest

- Usefulness of the generated visualizations
- Ease of command system use

Users generally found NLIs helpful: 72% rate charts as "Useful" or "Very Useful."

40% found the NLIs "Easy" or "Very Easy" to use, with an additional 22% feeling neutral.

Learning to use NLIs without prior instruction added to cognitive demands.

Lack of guidance required participants to adapt quickly while focusing on collaborative tasks with their study partners.

This highlights both the value and usability challenges of NLIs in exploratory data tasks, especially in collaborative settings.



Part - 2

Challenges for users → Clarification
Requests

Challenges observed from interactions during the EDA Tasks..



Difficulty in understanding generated visualization



"Can you show me the uninsured rate for different counties?"



Generates a grouped bar chart of uninsured rate, grouping by county-type



"I don't understand very well, what these bar charts are for?"



"So, it is grouping them by county type."



Difficulty in recovering spoken information



"We'll discuss the comparison of the graph, we can compare it and generate the resources for every disease that we have."



"Sorry, I didn't get you. So, we already have this COVID?"



"For the COVID, we can come to a conclusion that. . ."



Uncertainty about system capability and procedure



"Can we just mix and match everything and try it?"



"Yeah, I think we need to mix and match two things."



"Can we do more than two?"



No, I think only two different factors are involved here."

Clarification Requests (CR)



Seeking and providing clarifications is central to human-human communication



Speakers rely on CRs to negotiate meaning, resolve misunderstandings, and coordinate subsequent actions.



Grounded Clarifications^[12] involve CRs tied to specific, real-world contexts or modalities, ensuring mutual understanding between the participants in a conversation



For ***utterance U***, subsequent turn is considered ***grounded clarification in modality m*** (e.g., visual, auditory) if there is lack of positive evidence of understanding in that modality

[12]Luciana Benotti and Patrick Blackburn. "A recipe for annotating grounded clarifications". In: Proceedings of the 2021 NAACL

Grounding of Clarification Requests (CR) observed in recorded transcripts of Study 1...

Visual Modality



Difficulty in understanding generated visualization

 "Can you show me the uninsured rate for different counties?"

 Generates a grouped bar chart of uninsured rate, grouping by county-type

 "I don't understand very well, what these bar charts are for?"

 "So, it is grouping them by county type."

Auditory Modality



Difficulty in recovering spoken information

 "We'll discuss the comparison of the graph, we can compare it and generate the resources for every disease that we have."

 "Sorry, I didn't get you. So, we already have this COVID?"

 "For the COVID, we can come to a conclusion that. . ."

Kinesthetic Modality



Uncertainty about system capability and procedure

 "Can we just mix and match everything and try it?"

 "Yeah, I think we need to mix and match two things."

 "Can we do more than two?"

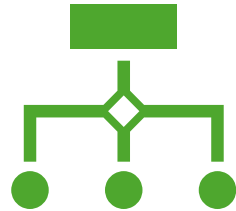
 "No, I think only two different factors are involved here."

Existing CR Datasets..



Existing datasets
(SCARE) ^[12] show this
grounding can be
annotated

But they do **not capture**
real-world
collaborative task
performance

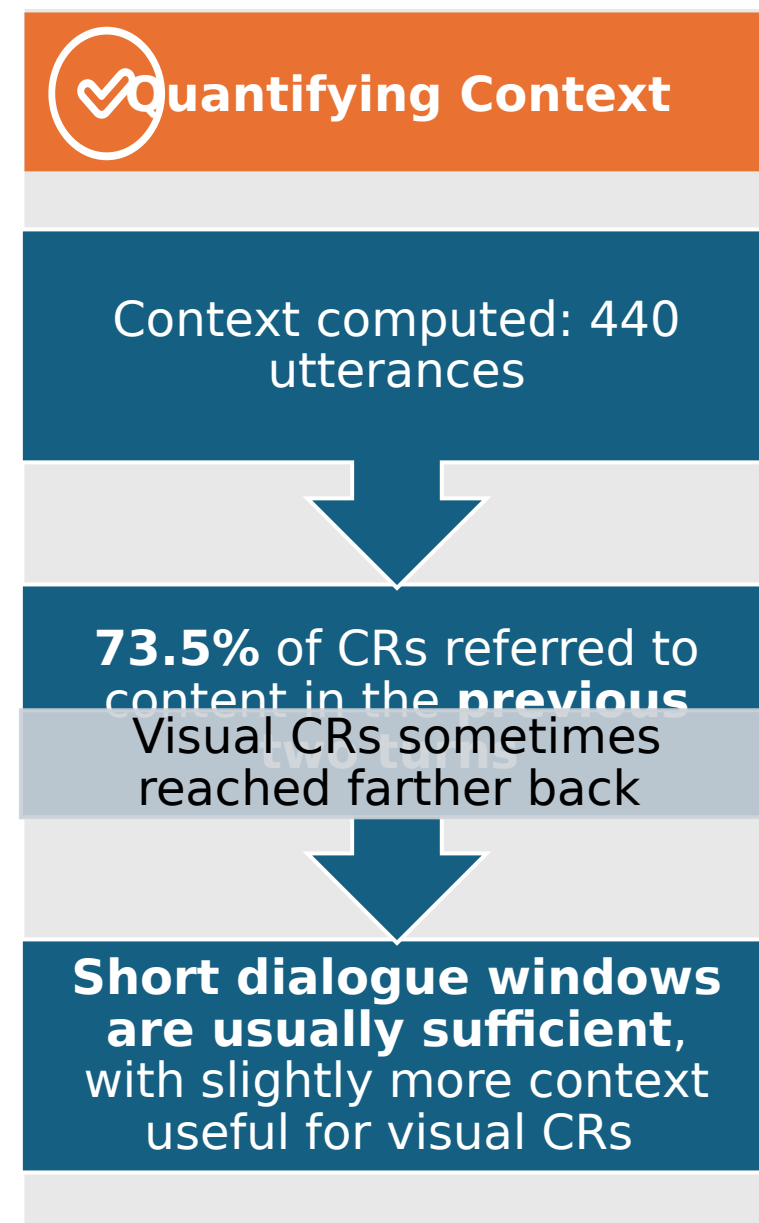
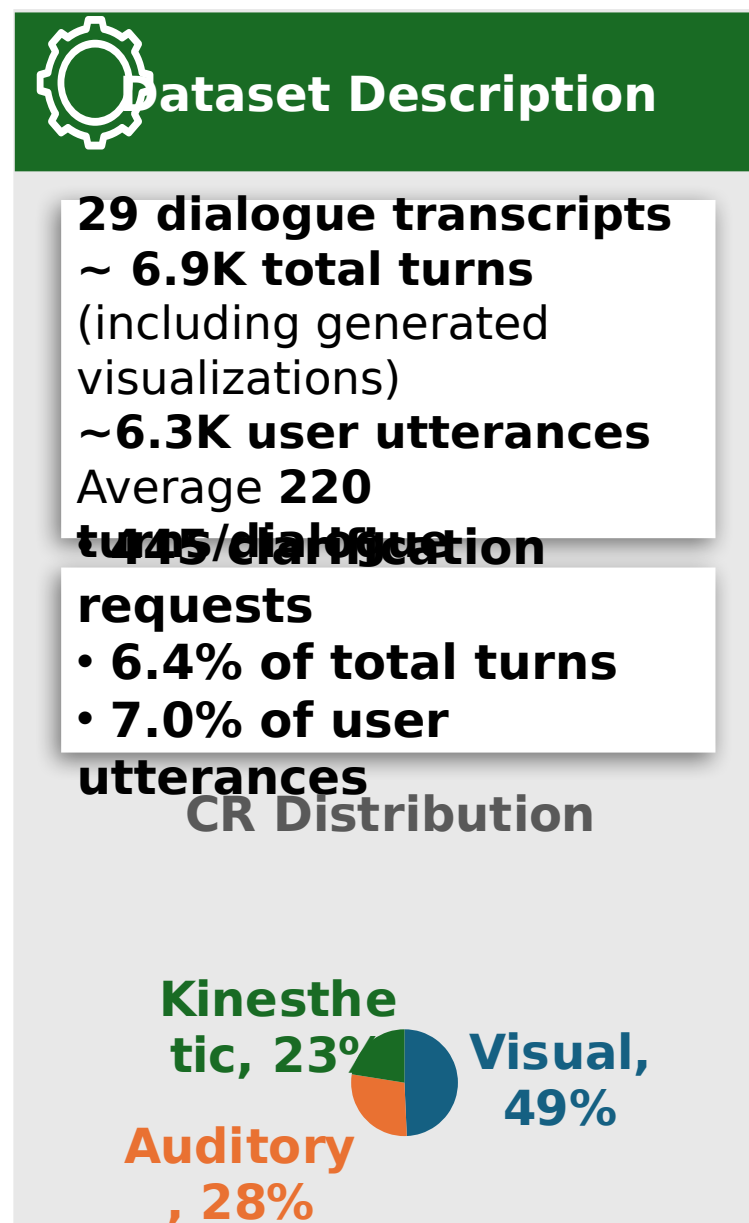
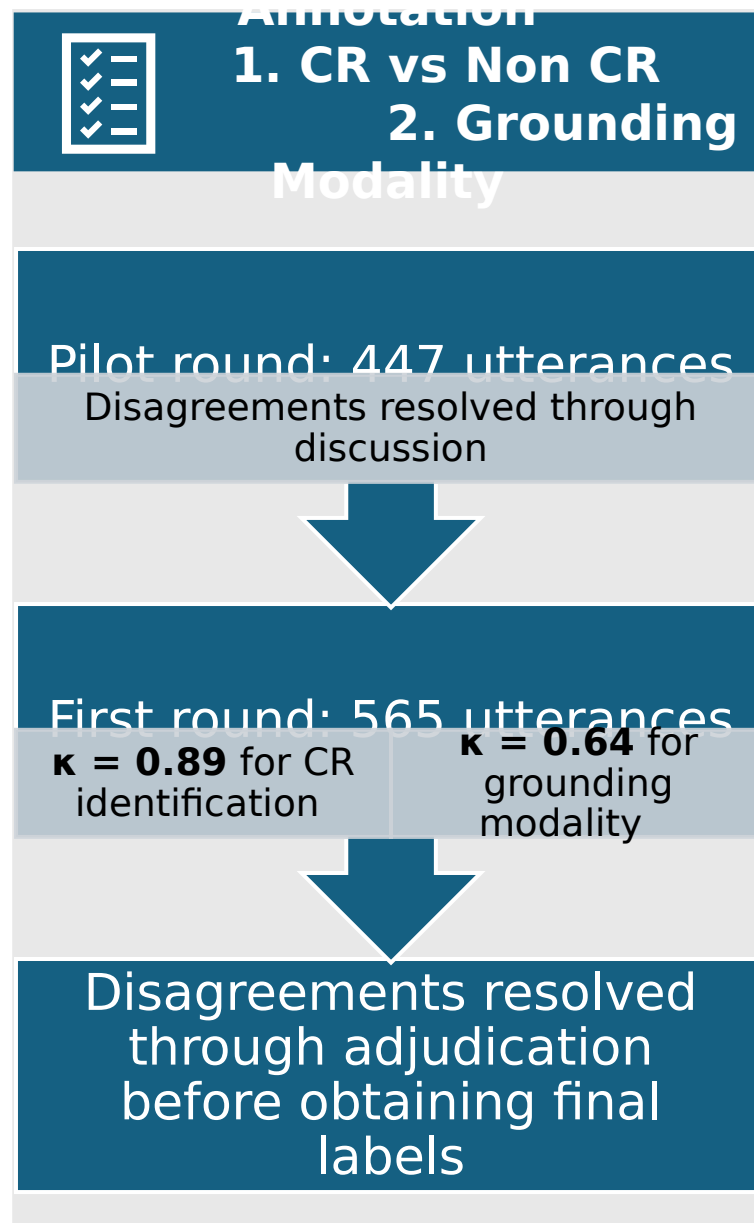


Other CR datasets are
also:

- asynchronous
- single-turn
- not built for collaborative task-oriented dialogue

**We need clarification datasets grounded in
real-time collaborative task interaction**

ClarVis: Annotation, Dataset Description and Context



Baseline Experiments using ClarVis dataset

Best CR-ID model

- **DistilBERT + Prev1**
 - CR F1 = **0.66**
 - CR Precision / Recall = **0.75 / 0.59**

Best GMOD model

- **TF-IDF + Logistic Regression**
 - Visual F1 = **0.73 (+ Prev1)**
 - Kinesthetic F1 = **0.55 (no context)**
- **DistilBERT**
 - **Auditory F1 = 0.43 (no context)**

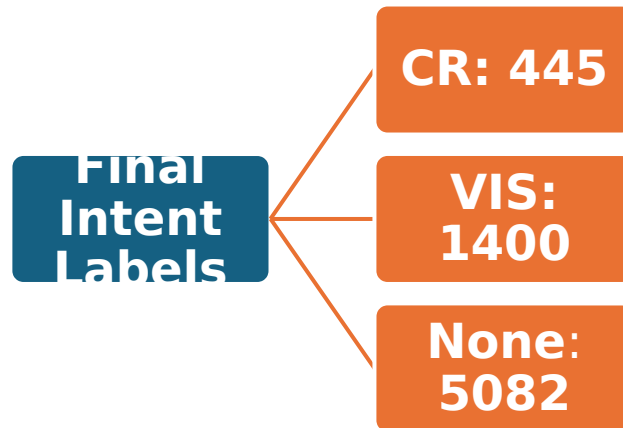
Overall baseline pattern on grounding modality classification: Visual > Kinesthetic > Auditory

Takeaways

- Baselines show that ClarVis contains learnable signals for both CR identification and grounding modality classification.
- Short dialogue context helped CR-ID, especially with DistilBERT.
- Visual grounding: most recoverable modality

Labelling Visualization Requests(VIS)

- **Non-CRs in 565 utterances labeled for 'VIS'**
- **$k = 0.91$**



Focus on Visual CRs

Auditory CRs: ❌

- Gaps in hearing
- Directed to the co-participant

Kinesthetic CRs: ❌

- How to interact with the system
- ~~Operate at a more meta-level~~

Visual CRs: ✅

- Tied to system's core functionality
- Concern interpreting generated visualizations

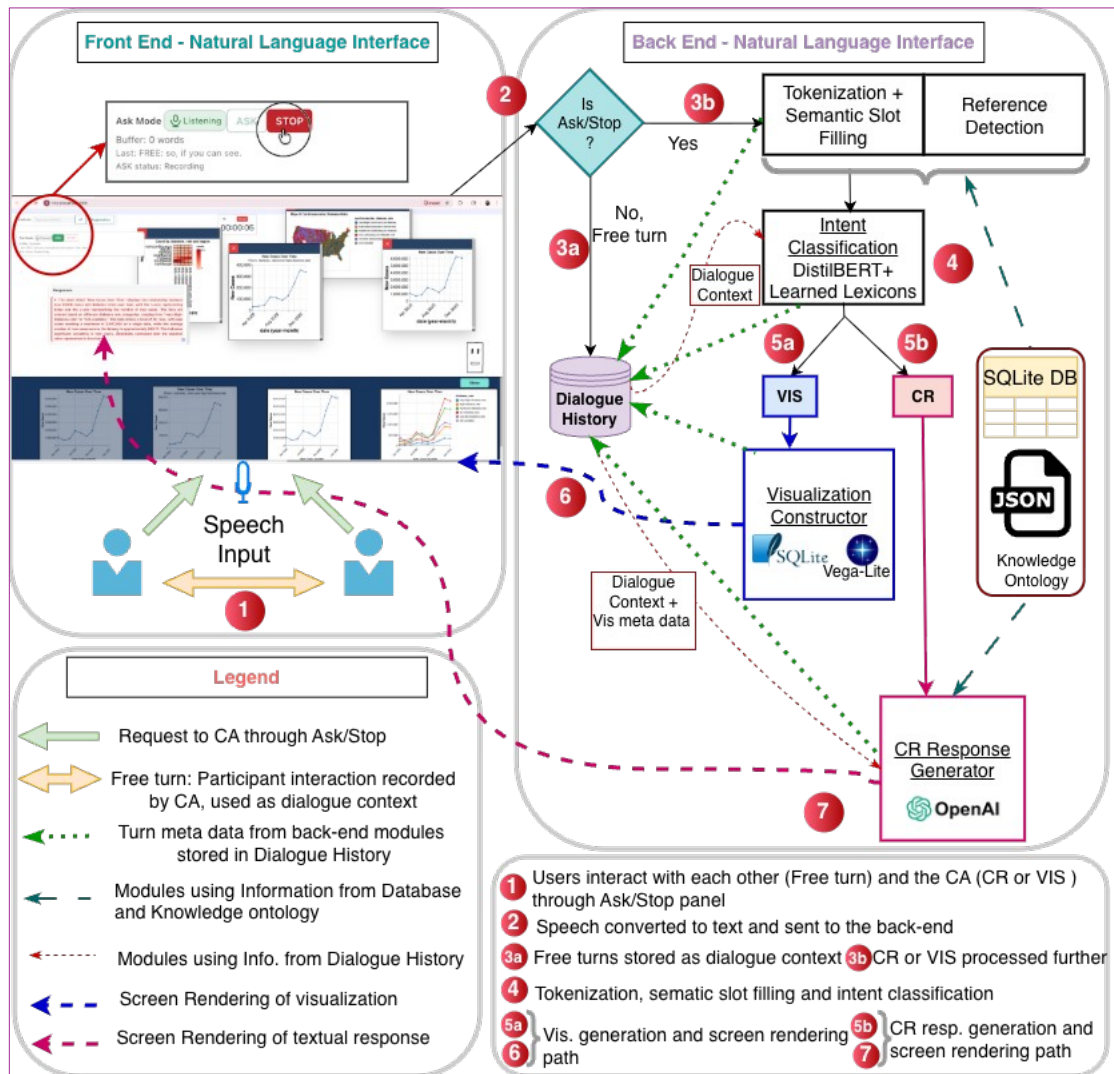


Part - 3

CRs in updated CA

3A. Developing Clarification-Aware Conversational Assistant and User Study Design (Study 2)

A Clarification-Aware Conversational Assistant (CA) Pipeline for Collaborative Data Exploration



Why the pipeline architecture?

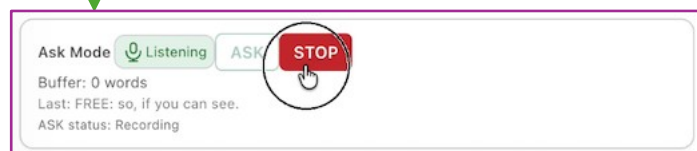
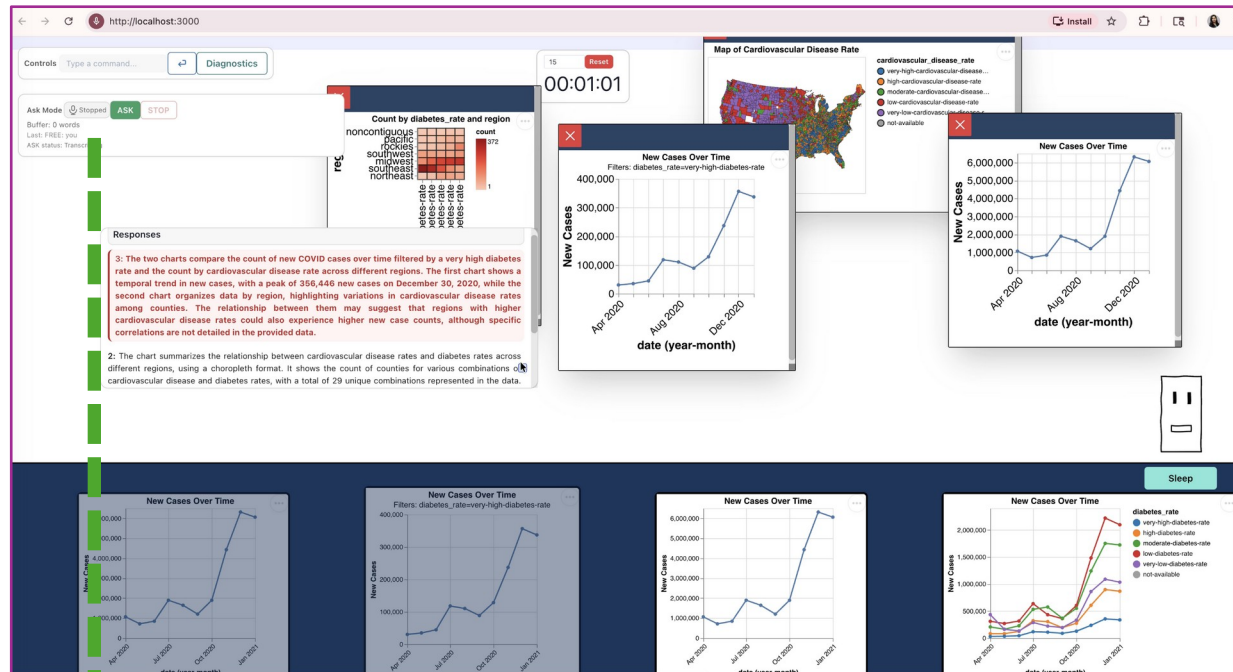
Multiple subtasks of interest:

- intent detection
- reference resolution
- visualization construction and generation
- clarification response generation.

More controllable and

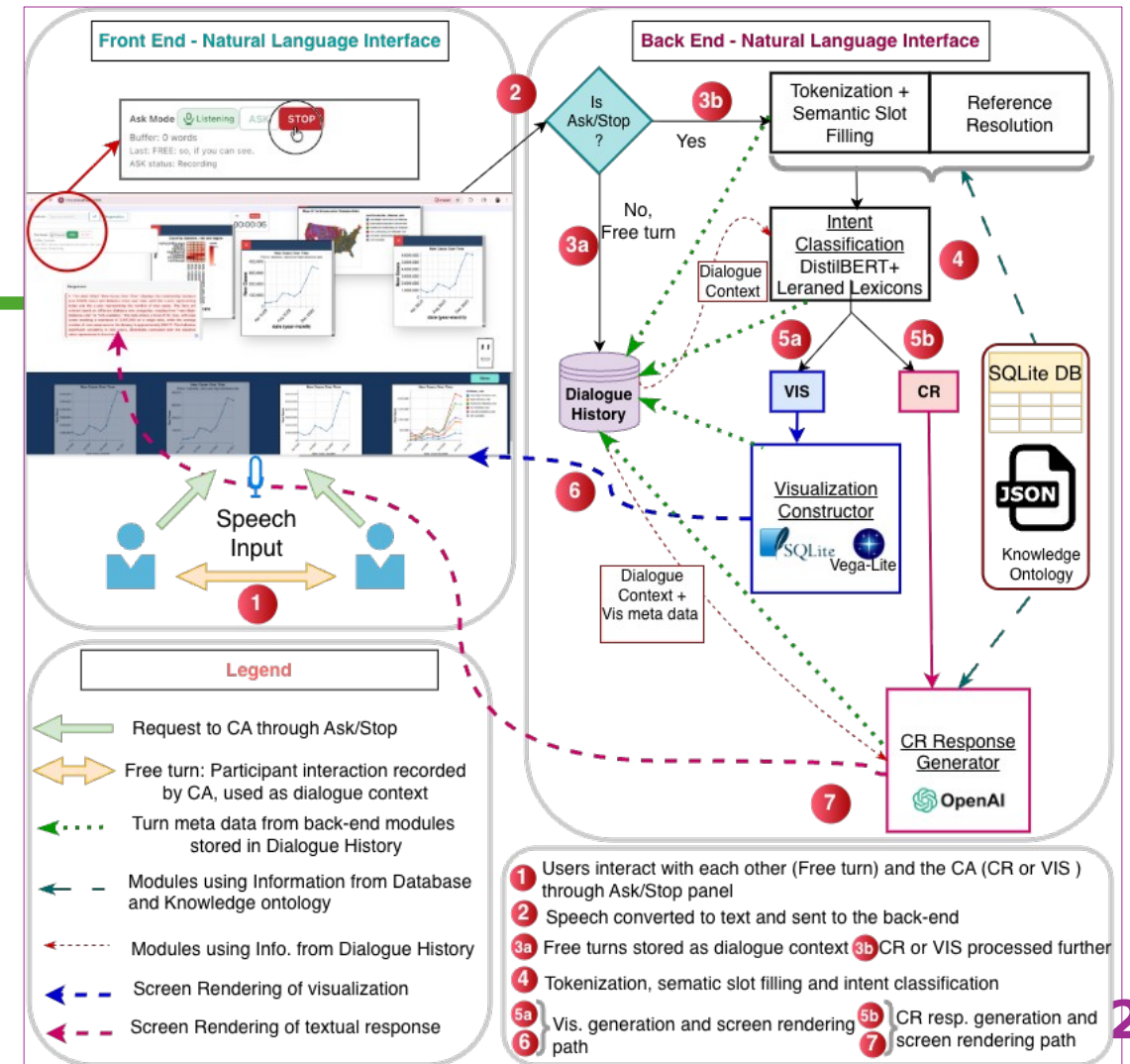
- Each stage can be **debugged, evaluated, and improved separately** while keeping outputs **grounded in the data.**

A Clarification-Aware Conversational Assistant (CA) Pipeline for Collaborative Data Exploration : Natural Language Interface

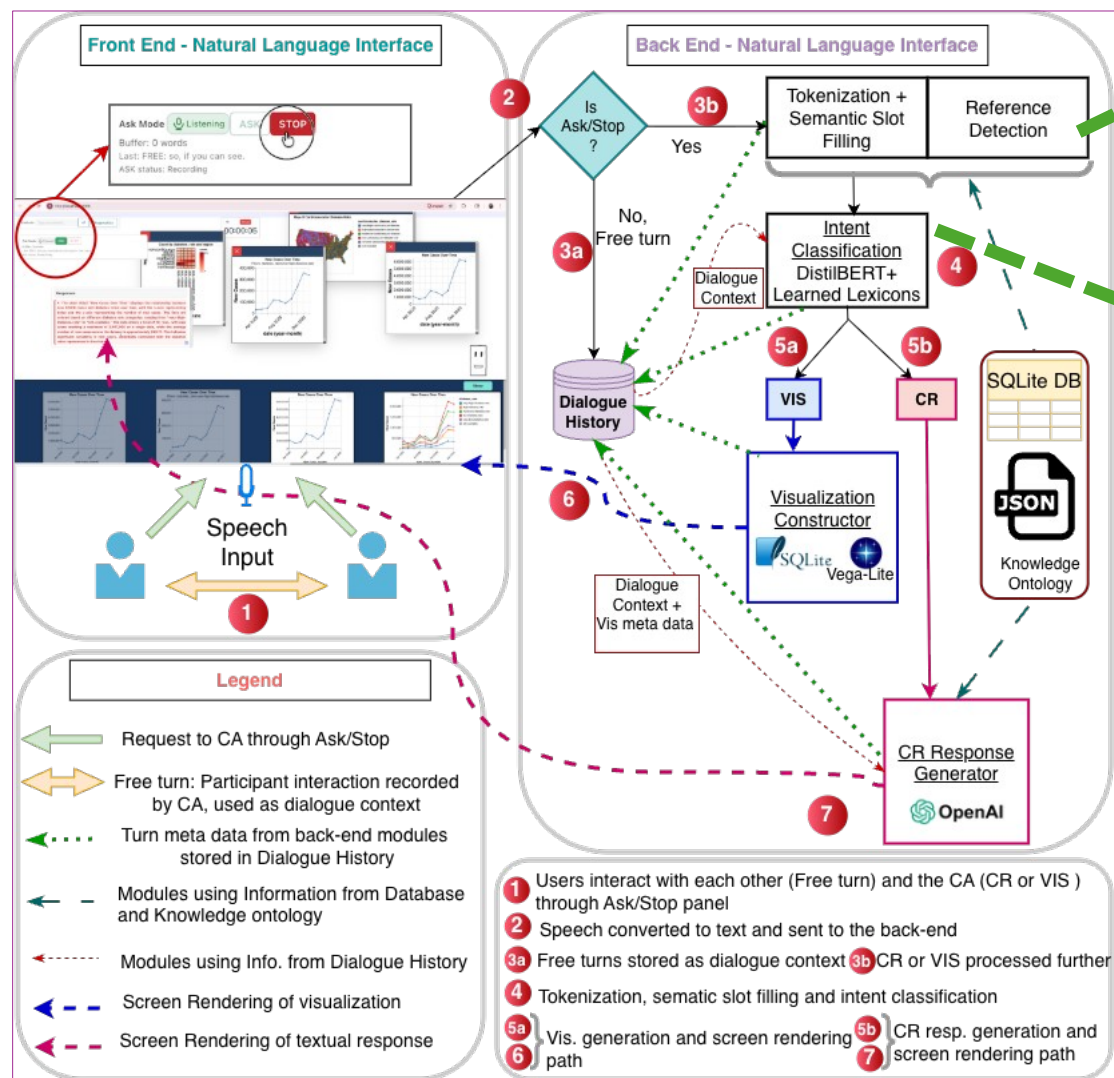


Two speech input modes:

- Ask/Stop: Directed to the CA
- Free: Directed to the



Back- End Components - 1



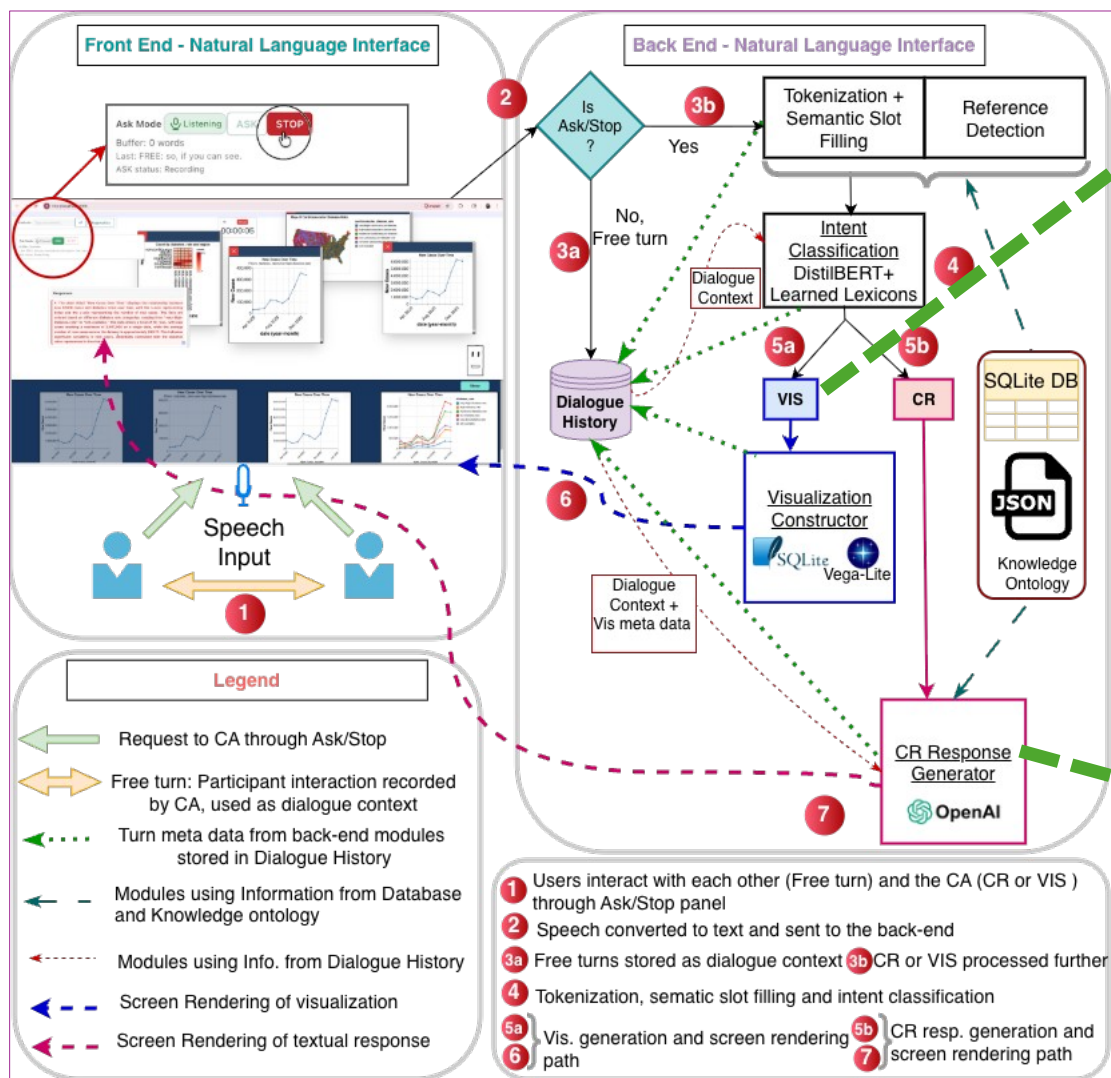
Tokenization and Semantic Slot Filling:
“Cardiac ailments <cardiovascular disease rate> in the west coast <region: pacific>”

Reference Detection: “Can you explain what exactly each color represents in this chart?”

Intent Classification (CR, VIS, None)

- DistilBERT (focal loss)+ Prev3 (trained on ClarVis)
 - CR(75.0/73.1/74.0)
 - VIS (79.2/81.0/80.1)
 - None(92.5/92.1/92.3)
- Learned Lexicons for Visual Modality CR (CR-V)
 - Symbolic post-processing override for low confidence (Prob. <0.55) Intent model prediction
 - 394 CR-V phrases learned from training data: ≥ 3 occurrences and $\geq 55\%$ CR-V association.

Back- End Components - 2



VIS branch

Resolved semantic slots → **Chart type selection**
→ **SQL query construction** → **Vega-Lite spec generation**

CR branch

Prompting strategy:

- Structured evidence-grounded prompting
- gpt-4o-mini
- System prompt (fixed) + User prompt (dynamic runtime context)

Prompt:

- System:** CR type (explanation / chart element meaning / comparison) → generate grounded answer from ordered chart/data evidence
- User:** utterance + reference chart + recent turns + resolved slots + supporting facts/statistics

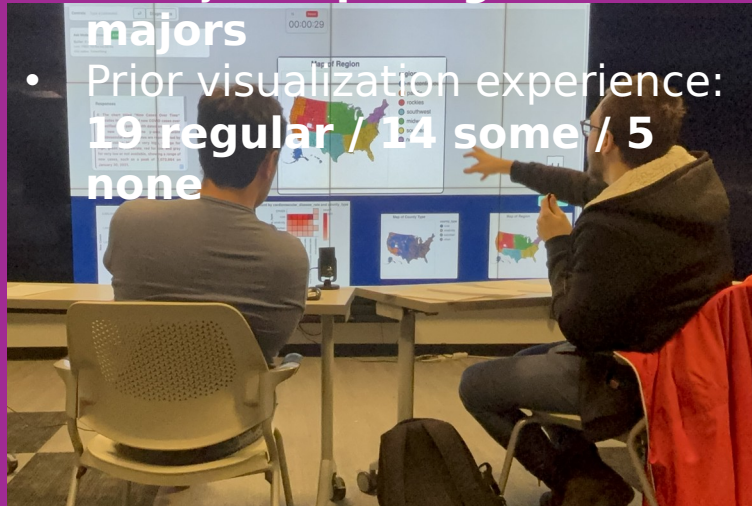
A Clarification-Aware Conversational Assistant (CA) Pipeline for Collaborative Data Exploration : User Study Design

Exploratory single-condition study:

Understand how users used the clarification-aware CA in practice

- 38 participants → 19 Groups
- No overlap with earlier user studies
- Age 18-29 (M = 22.82, SD = 3.01)
- Mostly computing-related majors

- Prior visualization experience: 19 regular / 14 some / 5 none



Design choices

- Encouraged CRs: framed as “questions about data visualizations”
- Task structure: 5-min warm-up + 3 x15-min tasks
- Open ended EDA tasks
- Seed questions / examples: To support

Participant instructions

- Introduced system as **research prototype**
- Work **collaboratively**
- Use **charts + textual responses to visualization-related questions as evidence**
- Rely only on **provided COVID-19 dataset**
- Speak clearly, stay task-focused

All sessions were Audio, Video and Screen Recorded.
Session Logs from the code were also collected for analysis.

3B. Analysis of Data Collected from User Study (Study 2) Logs

Analysis of Data Collected From the User Study : Interaction Profile through CR & VIS

Statistic	Total	CR	VIS
Total	7690	235	653
Overall rate	–	3.06	8.49
Mean per group per task	103.92	3.18	8.82
Median per group per task	107.00	3.00	9.00
SD per group per task	49.46	2.31	4.03
Min per group per task	11.00	0.00	2.00
Max per group per task	208.00	10.00	20.00
Mean per group	404.74	12.37	34.37
Median per group	382.00	13.00	35.00



Participants primarily engaged with the system as a tool for generating visual evidence.



CRs played a more secondary, supporting role



Aligns well with study design → core task was collaborative EDA rather than question answering alone.



Consistent with prior work → although CRs play an important role in grounding and repair → usually make up only a small share of utterances in dialogue corpora

Analysis of Data Collected From the User Study : Interaction Profile through CR & VIS

Task-wise event density (Free / VIS / CR)

Free VIS CR



- CRs was **episodic and not the same across all tasks**.
- Emerged at **moments of uncertainty or interpretation**
- Clarification responses possibly helped users **return to visualization generation** rather than remain in repair

Density patterns

- **Free turns:** spread more evenly across tasks ; **VIS** and **CR:** showed more localized peaks
- **Warm-up:** VIS spread across task; CR near the end
- **Task 1:** **VIS → CR → VIS** pattern
- **Task 2:** CR rose soon after an early VIS peak
- **Task 3:** CR appeared earlier, VIS later

What followed CR responses

- **CR → VIS: 146 / 235 (62.13%)**
- **CR → CR: 66 / 235 (28.09%)**
- **CR → Free: 23 / 235 (9.79%)**

Analysis of Data Collected From the User Study : System's Responses to CRs

Checklist-based LLM-as-a-judge Inspired by **CheckEval** - More **interpretable** and **stable** than one overall rating

4 dimensions: Correctness, Groundedness, Helpfulness, Clarity

Scoring:

- **5 yes/no checks per dimension**
- **20 binary checks per response**
- Score = % **yes** per dimension

Judges - prompted with the CR, Response and Dialogue Context along with the scoring rules

- **gpt-4.1-mini (note: used gpt-4o-mini to generate CR responses)**
- **gemini-2.5-flash-lite**
- **qwen2.5-14B-Instruct**

Humans & LLMs as judges

2 coders, 50 CRs: Cohen's κ

Correctness	0.64 7
Groundedness	0.54 0
Helpfulness	0.63 7
Clarity	0.35 0

LLM agreement with human gold standard (κ)

LLM Judge	Correctness	Groundedness	Helpfulness	Clarity
<i>Gemini 2.5 Flash Lite</i>	0.43	0.26	0.32	0.43
<i>GPT-4.1-mini</i>	0.44	0.25	0.52	0.28
<i>Qwen 2.5 14B Instruct</i>	0.30	0.08	0.23	0.28

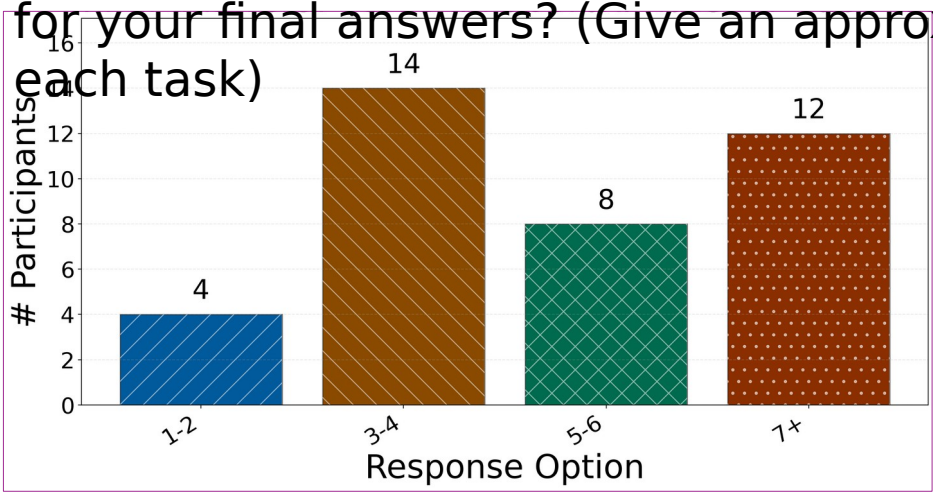
Final Assessment of responses

Dimension	Selected judge	Agreement κ	Final score %
Correctness	<i>GPT-4.1-mini</i>	0.44	87.2
Groundedness	<i>Gemini 2.5 Flash Lite</i>	0.26	89.6
Helpfulness	<i>GPT-4.1-mini</i>	0.52	79.2
Clarity	<i>Gemini 2.5 Flash Lite</i>	0.43	98.4

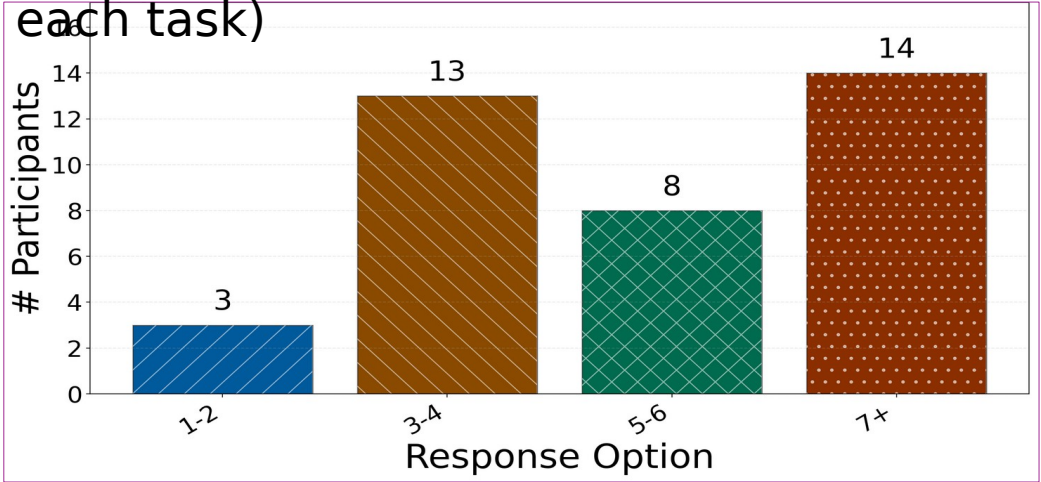
3D. Analysis of Data Collected from User Study (Study 2): Post Study Questionnaire

Post Study Questionnaire : Chart Use, Visualization-Related Questions, and Perceived Usefulness

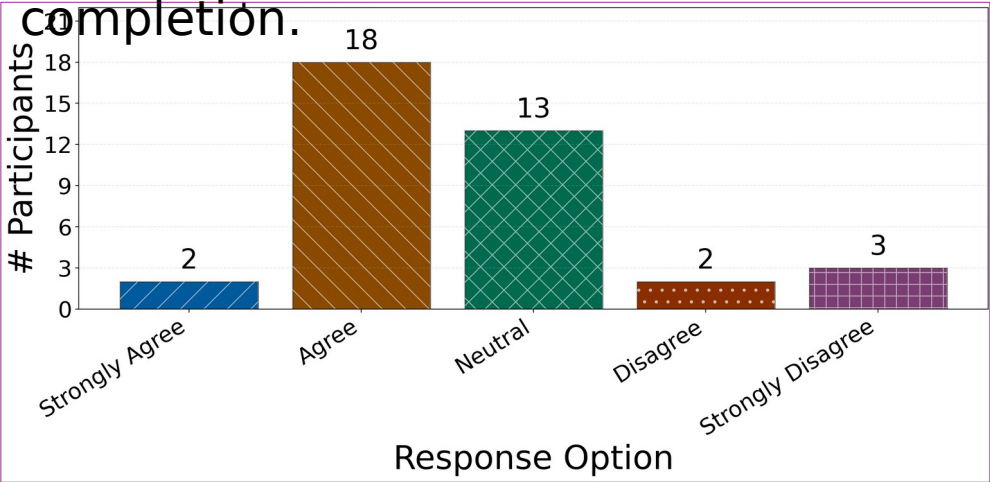
How many charts did you actively use as evidence for your final answers? (Give an approximate for each task)



How many visualization-related questions did you ask? (Give an approximate for each task)

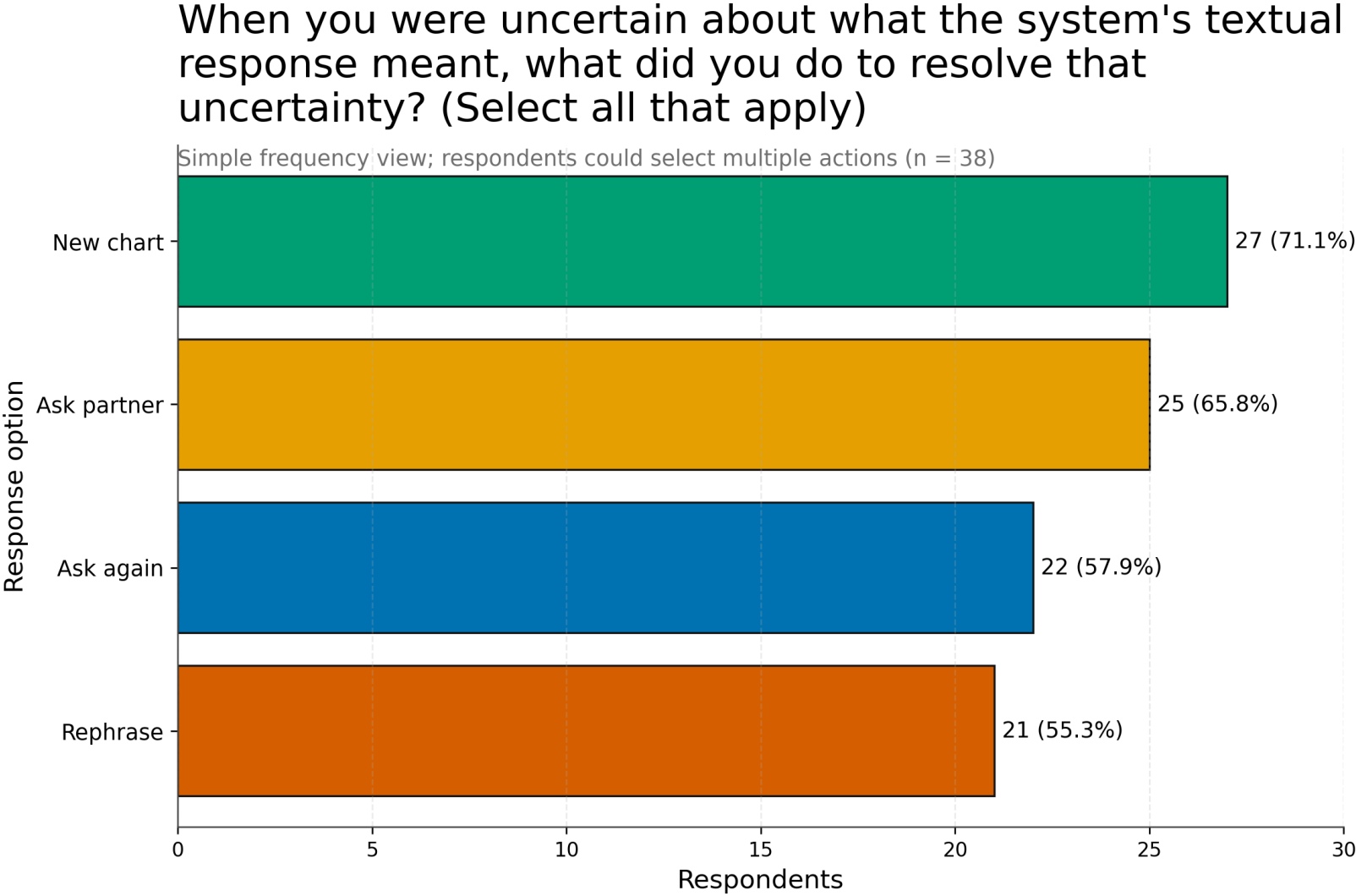


The system's textual responses to your questions were useful for task completion.



Significantly associated

Post Study Questionnaire : Participant-Reported Response Handling



Conclusion: Why This Research Direction Matters?



AI-assisted visual analysis is growing

Recent systems support AI assistance, GenAI-based visual analytic conversations, iterative visualization creation, and more complex text-to-visualization outputs.



But user experience is still underexplored

We need more evidence on how users work with CA support during analysis:
how they interpret outputs, handle uncertainty, and continue the task.



This matters beyond one system

As data-driven decision-making becomes more common, people need tools that help them make sense of data collaboratively.



This dissertation points to one path forward

A clarification-aware CA can support collaborative data exploration by combining:
accessibility + visualization + clarification support



Broader takeaway

The full goal is not yet achieved, but this work shows that natural, clarification-aware, collaborative CAs are a meaningful direction for making data exploration more accessible, interactive, and useful.

Conclusion: Future Extensions



Other datasets and EDA tasks

Test the clarification-aware CA with different datasets and exploratory analysis tasks.



Larger collaborative groups

Study how the CA supports groups with more than two users, where coordination and shared understanding may be more complex.



Less constrained data-exploration settings

Evaluate the CA in longer and more varied collaborative settings, such as classroom, workplace-style planning, or public-health/policy analysis tasks.



Other AI-assisted data tools

Apply the clarification-focused analysis to dashboard, spreadsheet, or report-generation assistants, where users may also need to question, verify, or repair AI-generated outputs.

Appendix

Natural Human-Human Conversations are Characterized by ...

1	User 1: I think we can do Covid risk first, right?
2	User 2: We have a very high Covid rank.
3	User 2: What do you think about it?
4	User 1: Should we get a map of the county type first?
5	User 1: Actually, it doesn't say that, well, what is the ongoing problem right now in America?
6	User 2: The first is poverty, right?
7	User 2: So, I think, wait.
8	User 1: Well, I think we can do like Covid risk and then access to doctors and then...

References to prior queries to accommodate the contextual information

Ambiguous utterances

Incomplete queries

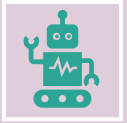
To explore **how** addressing the identified challenges faced by users in Study 1 shape user interaction with an updated CA, I ask the following Research Questions..



RQ4: How do the system's responses to user-initiated clarification requests relate to the **interaction flow** and **task progress** during collaborative data exploration?



RQ5: How do **users perceive the usefulness and interpretability** of these responses for the EDA task completion?



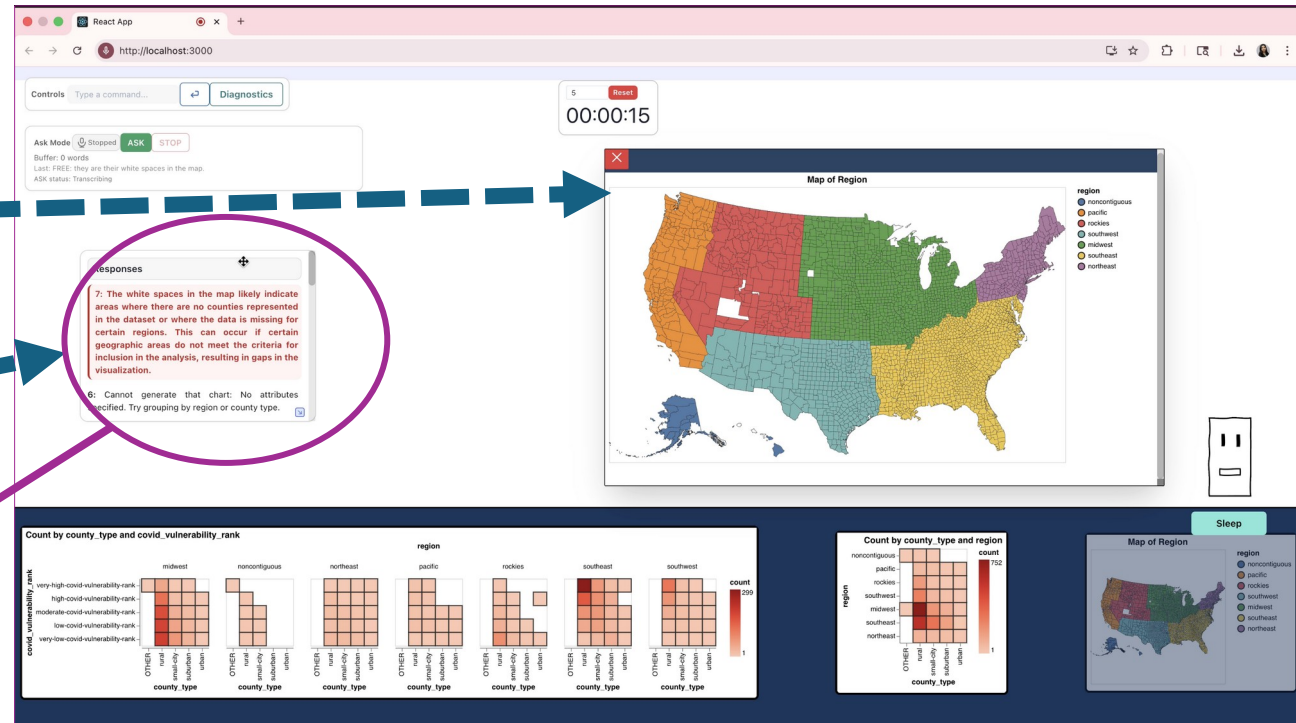
Update existing CA framework so that it identifies user-initiated CRs and responds to those CRs



Design a user study with the updated CA to answer RQ4 and RQ5

A Clarification-Aware Conversational Assistant (CA) Pipeline for Collaborative Data Exploration – Sample Conversation

Group 21 Warm Up Task			
17	Give me a bar chart for that	User 1	VIS
18	by the region	User 1	VIS
19	they are their white spaces in the map.	User 2	Free
20	Why are there white spaces in the map?	User 1	CR



Responses

7: The white spaces in the map likely indicate areas where there are no counties represented in the dataset or where the data is missing for certain regions. This can occur if certain geographic areas do not meet the criteria for inclusion in the analysis, resulting in gaps in the visualization.

A Clarification-Aware Conversational Assistant (CA) Pipeline for Collaborative Data Exploration : Charts Types

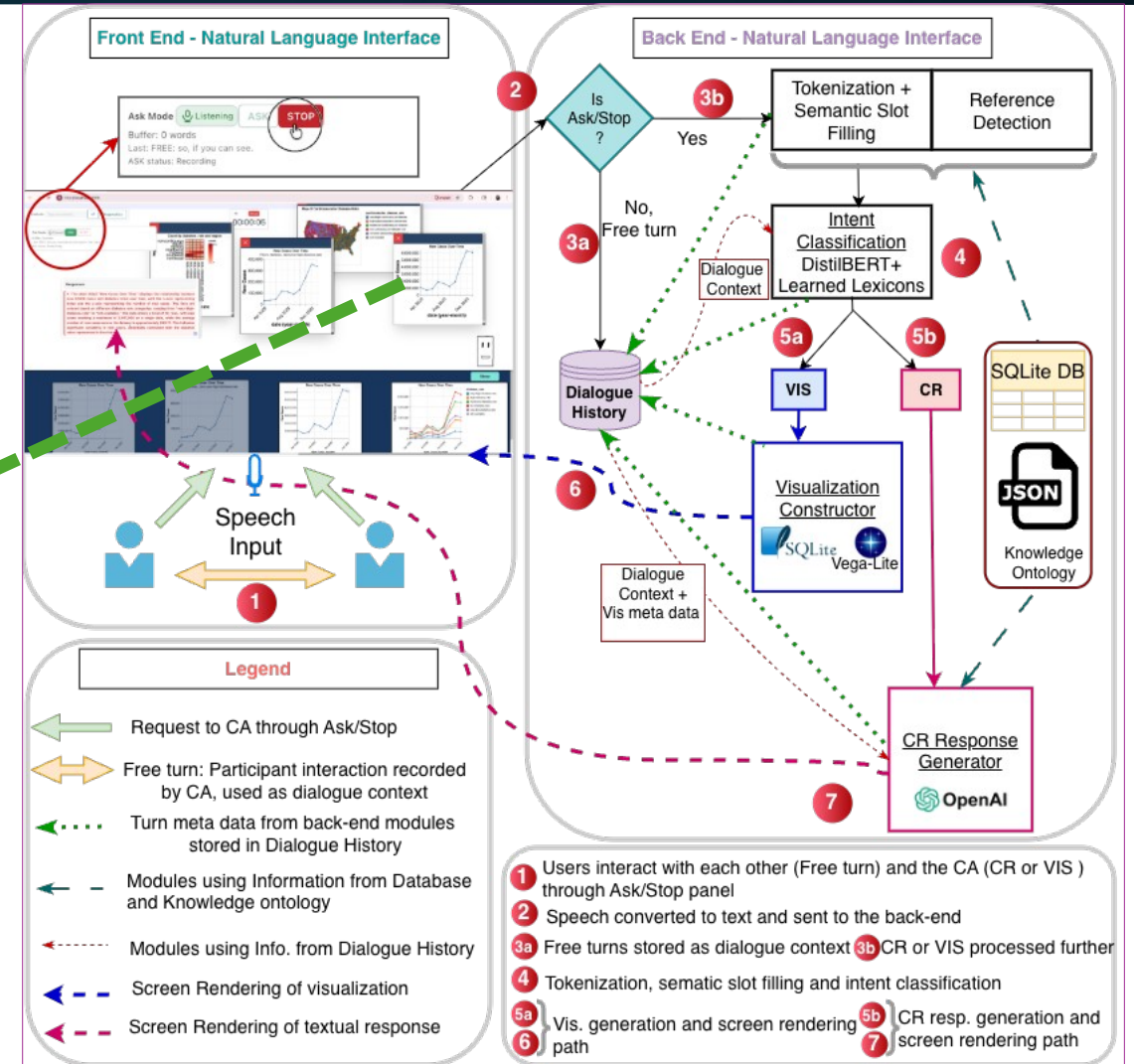
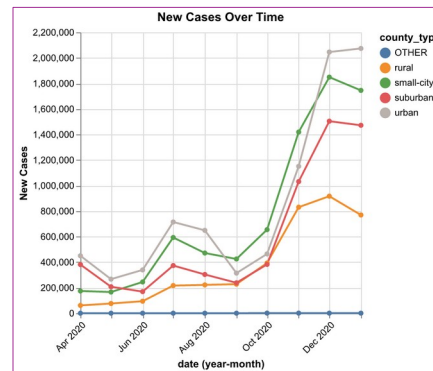
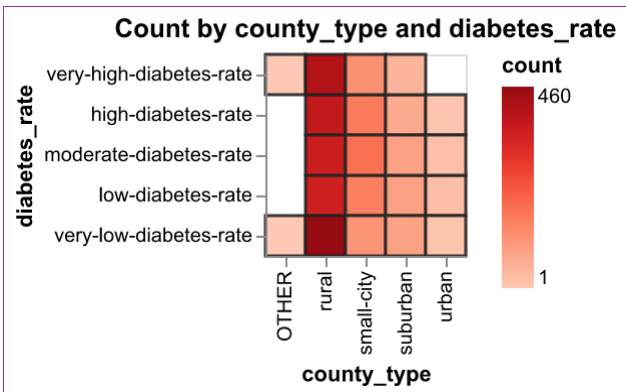
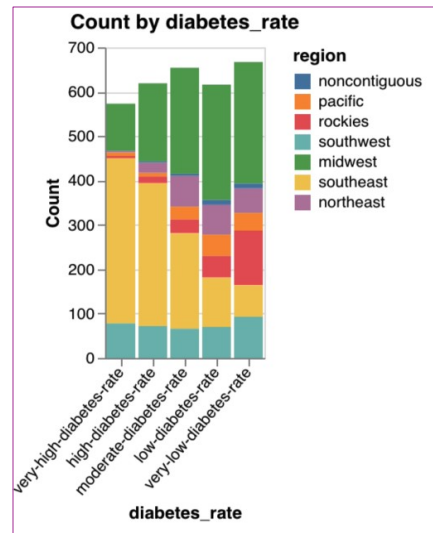
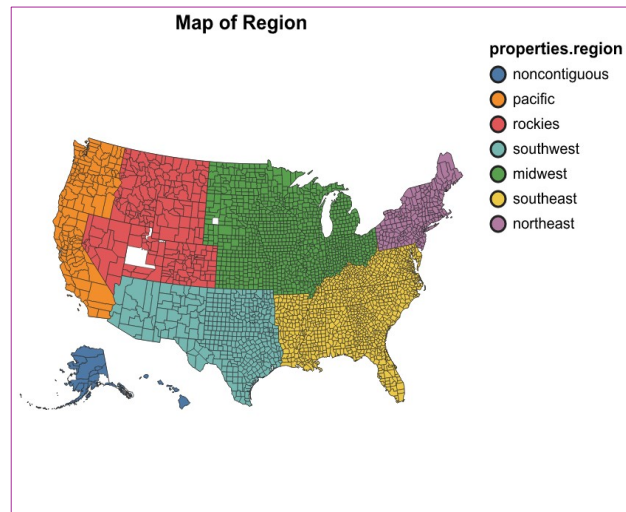
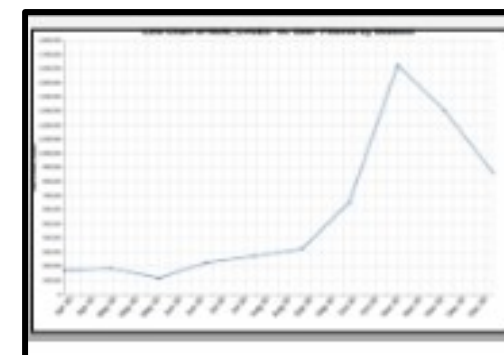
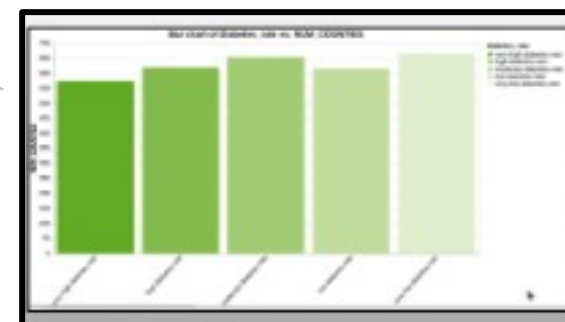
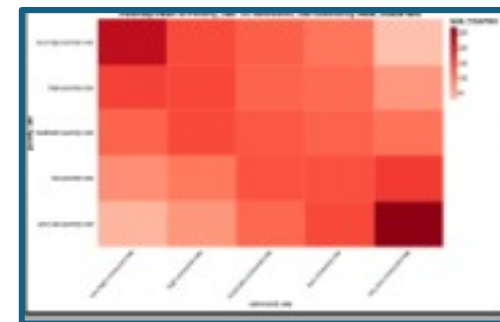
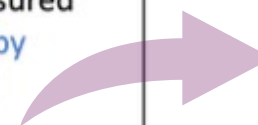


Chart Type	Data (Essential)	Data (Optional)	Example
<i>Choropleth Map</i>	Nominal Attribute X FIPS (Federal Information Processing System) Code	Nominal Attribute Value	Choropleth Map of Diabetes Risk (Filtered by Very High Covid Vulnerability Rank)
<i>Heatmap</i>	Nominal Attribute (X-Axis) X Nominal Attribute (Y- Axis)	Nominal Attribute Value	Heatmap of Poverty Rate and Uninsured Rate (Filtered by Southeast and Southwest)
<i>Bar Chart</i>	Nominal Attribute (X-Axis) X Quantitative Attribute (Y-Axis)	Nominal Attribute Value	Bar Chart of Diabetes Risk vs Number of Counties (Filtered by Midwest)
<i>Line Chart</i>	Temporal Attribute(X-Axis) X Quantitative Attribute (Y-Axis)	Nominal Attribute Value	Line Chart of Covid Cases over Time Span of 10 months (Filtered by Very Low Cardiovascular Disease Rate)



Data Collected From the User Study: Open Ended Task Conclusion Analysis - Group-Task Aggregation

Why aggregate at the group-task level?

- Tasks were completed collaboratively
- Each **group-task** as the unit of analysis.
- The coded final task conclusions were treated as indicators of progress toward a **task-closing interpretation**, not as correctness scores, because the EDA tasks were open-ended.

Aggregation rule:

For each group-task → **higher score** across the two participants.

Why this rule?

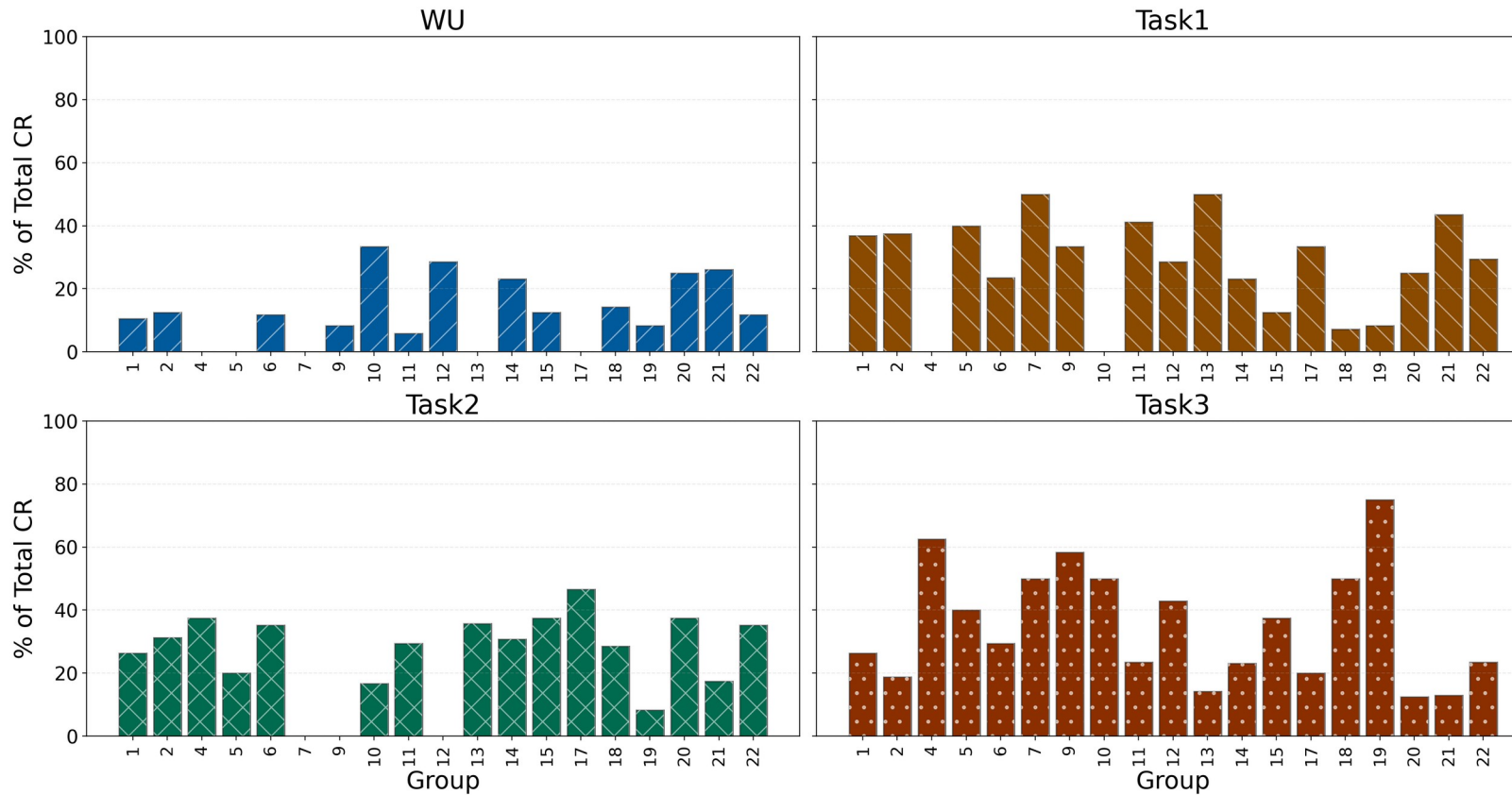
This captures whether a clear conclusion, justification, or use of evidence appeared anywhere in the group’s final interpretation.

Dimension	Participant-level coding
Conclusion presence	1 = no/weak, 2 = partial, 3 = clear
Justification	1 = none, 2 = some, 3 = elaborated
Evidence use	1 = none, 2 = any evidence used

Main task-progress takeaway

- After Warm-up, groups more often ended with recognizable conclusions, frequently accompanied by at least some justification and evidence use.
- This suggests that participants were generally able to work toward plausible task-closing interpretations once they moved beyond the initial orientation phase.

Analysis of Data Collected From the User Study : Interaction Profile through CR & VIS



Within-group percentage distribution of CR-s across tasks

- **Warm-up:** smallest avg. CR share **12.21%** → **exploration-heavy onboarding phase**
- **Task 1:** **30.78%**
- **Task 2:** **24.96%**
- **Task 3:** largest avg. CR share **35.30%**

Task-level concentration shows where CRs became more prominent → the next step is to see how they unfolded within the interaction itself.

3C. Analysis of Data Collected from User Study (Study 2) :Open-ended EDA Task Conclusions

Open Ended Task Conclusion Analysis

Each Participant's articulated task-closing interpretations were characterized along **5 dimensions**:

- **conclusion presence**
- **coherence**
- **justification**
- **evidence use**
- **uncertainty**

Task-wise Pattern (%participants in task conclusion categories)

	Warm-Up	Tasks 1-3
Clear conclusions	36.8%	64.7-68.4%
Any justification	36.8%	57.9-70.6%
Any Evidence	47.4%	55.3-79.4%

Reaching a conclusion and expressing it clearly were not the same

"Southeast.." (G7_T1_1), "Rural, southeast" (G9_WU_1)

Clear conclusion

"To answer the question, I think the insured rate and access to doctor move in the opposite direction" (G2_T3_2)

Justification and referring to evidence

"the heatmap shows the covid vulnerability is high.." (G4_WU_1)

Elaborated reasoning

"there are lot of people with more uninsurance ... it seems like it is moving in opposite directions"

Brief, tentative, or partially justified conclusion

"It is similar in southeast" (G5_T1_2)

Uncertainty

"we were confused ... so I am not sure" (G1_T1_1)

Open Ended Task Conclusion Analysis

After the warm-up, participants generally reached stronger, more justified, and more evidence-

Task 1: highest justification
and evidence use

Task 2: least uncertainty

Task 3: most clear conclusions
and elaborated justification



However, this broader progress was not explained by

CR/VIS frequency

averaged LLM-judge response-quality scores.



Overall, the findings suggest

Clarification responses mattered within collaborative EDA → mainly as part of local processes of verification, repair, and sense-making, rather than predictor of broader task outcomes

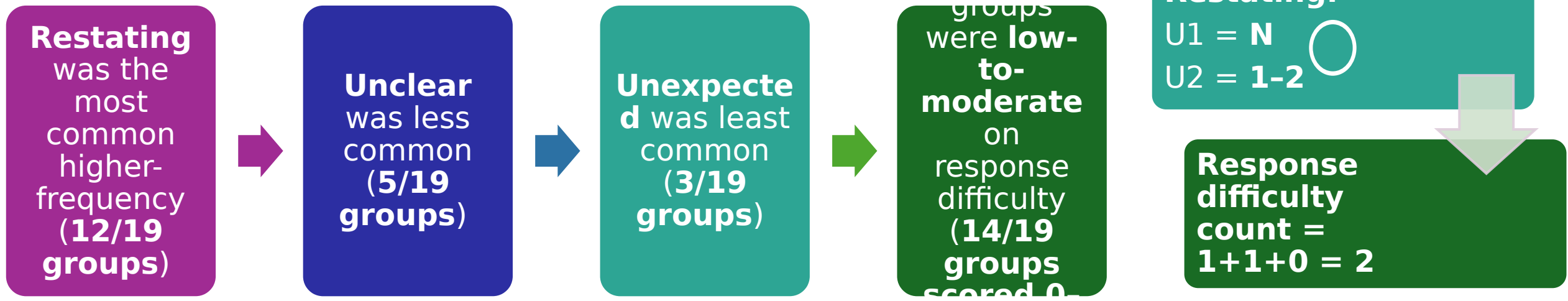
Post Study Questionnaire: Participant-Reported Response Handling: Group-Level Response Difficulty Variable

How was the variable derived?

For each difficulty type, a **group receives 1** if **either participant** selected a higher-frequency category (**3-4** or **5+**), and **0 otherwise**

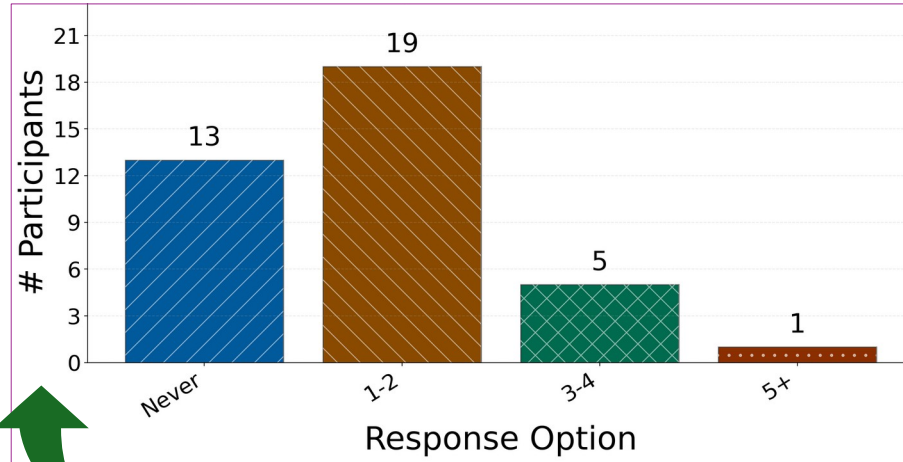
Score ranges from **0 to 3**.

This variable was used **only as an exploratory summary measure**.



Post Study Questionnaire: Participant-Reported Response Handling

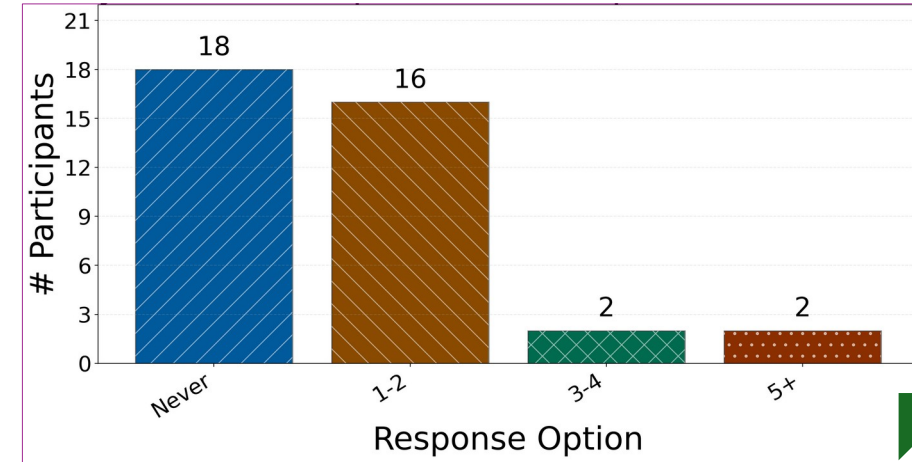
Were there times when the system produced a text response to your question but you did not know what it was referring to?



referential clarity → whether the response made sense after it was produced

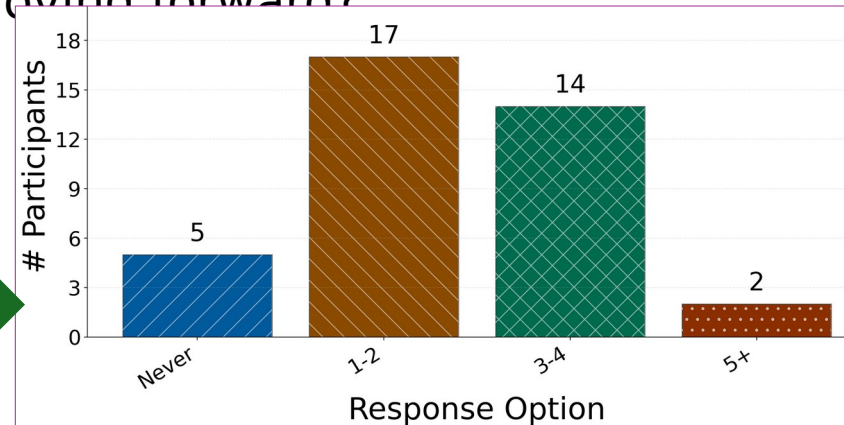
repair / reformulation indicator → participants had trouble with the

Were there times when the system produced a text response when you had not expected that response?



expectation match / mismatch → what users thought the system should do and what it actually produced

Were there times when you needed to restate the question multiple times before moving forward?



Post Study Questionnaire: Open Ended Questions

Reflexive thematic analysis: strategy and chart preferences

- What was your **overall strategy for completing the tasks**?
- What **kind of charts did you request** the most and why?

Theme 1: Iterative visual sense-making

Pattern	Participant s	Example quotes
Task decomposition and stepwise exploration	13 / 38	• “Starting with a basic chart and then asking questions further” (G2-U1)
Strategy refinement based on visual feedback	25 / 38	• “..try plotting common attributes and then find a common relation in the end between the plots you have generated” (G13-U2)

Takeaway:

Understanding was built gradually through successive visualizations, interpretation, and revision rather than through a single system response

Post Study Questionnaire : Open Ended Questions

Reflexive thematic analysis: strategy and chart preferences

- What was your **overall strategy for completing the tasks**?
- What **kind of charts did you request** the most and why?

Theme 2: Chart interpretability

Pattern	Participant s	Example quotes
Preference for relationship-revealing charts	15 / 38	• “Heat maps made sense to use for most of the questions to see if they correlate” (G1-U2)
Preference for visually straightforward charts	16 / 38	• “bar chart and as it is easy to find the maxima/minima at a glance” (G11-U2)

Takeaway:

Participants favored charts that supported quick pattern recognition and were easy to interpret.

Post Study Questionnaire : Open Ended Questions

Reflexive thematic analysis: response uptake and system interpretation

- Think about a visualization-related question you asked and the system’s response. What did you do after the system responded?
- What did you think the system’s response meant for your visualization?

Theme 3: Uptake of responses to clarification requests

Pattern	Participants	Example quotes
Verification of interpretation	15 / 38	• “We confirmed our observation with the system's response and got an explanation of the highest and lowest counts for each visualization” (G1-U1)
Extension of analysis	14 / 38	• “note down the extreme data points and try to correlate it with other data points from other data charts” (G6-U2)
Repair or selective uptake	9/38	• “I disregarded some of them. I used it as my answer for one” (G20-U1)

Takeaway:
Responses to clarification requests were actively taken up as part of the data analysis process

Post Study Questionnaire: Open Ended Questions

Reflexive thematic analysis: response uptake and system interpretation

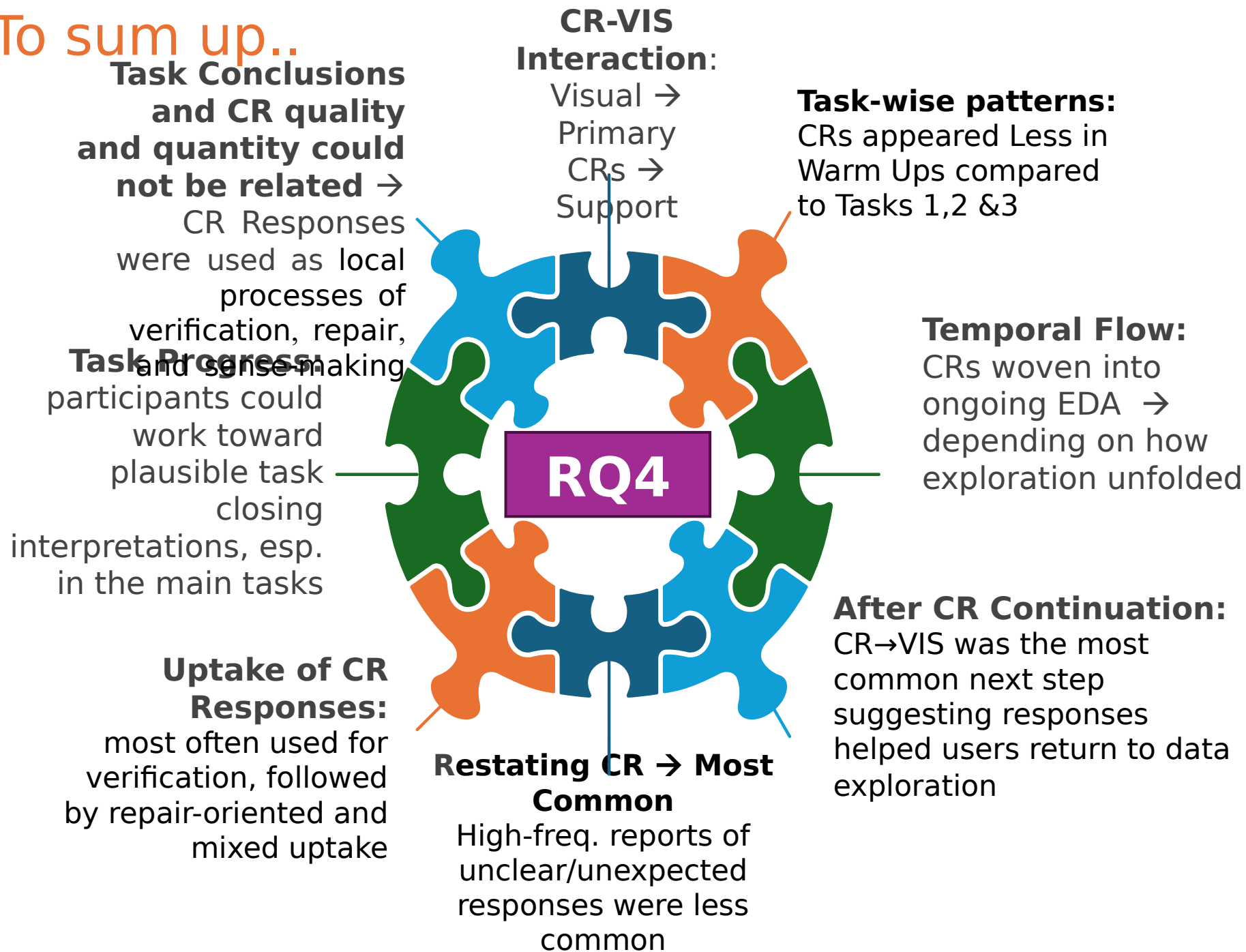
- Think about a visualization-related question you asked and the system’s response. What did you do after the system responded?
- What did you think about the system’s response?

Theme 4: Participant interpretation of how the system answered questions		
Pattern	Participants	Example quotes
Data-grounded explanations	21 / 38	• “The data base” (G12-U2), • “data it uses to generate the chart” (G14-U2)
Process-oriented explanations	14 / 38	• “Keywords from the speech and data and codes used to create the chart” (G12-U1).
Broad AI-based or uncertain explanations	9/38	• “Machine learning? I’m not sure, AI is a black box”(G10-U1)

Takeaway:

Participants generally understood the system as being grounded the dataset and generated visualizations, while also associating it with broader AI or language processing mechanisms

To sum up..

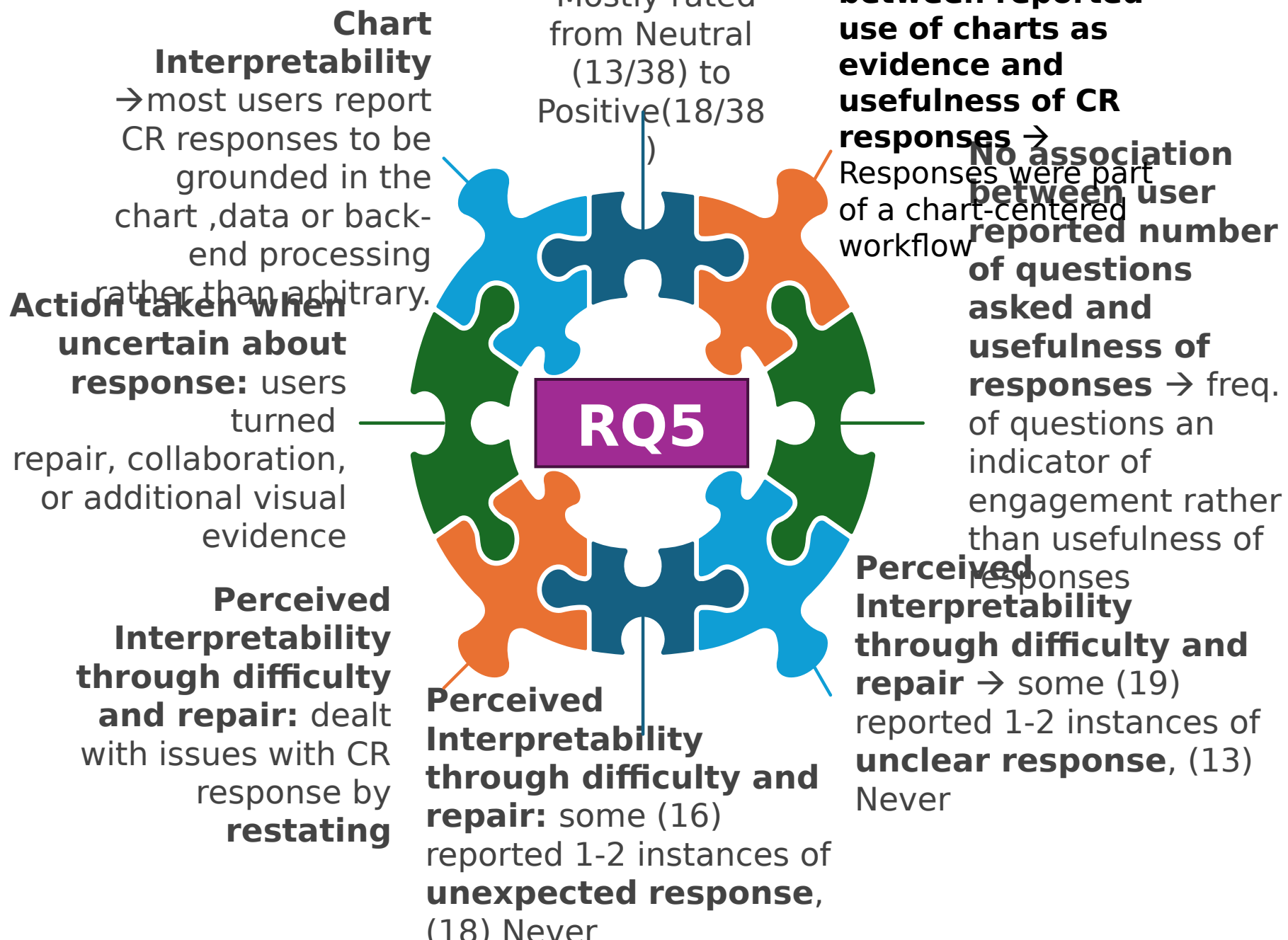


RQ4: How do the system's responses to user-initiated clarification requests relate to the **interaction flow** and **task progress** during collaborative data exploration?

Interaction Flow: How smoothly the collaborative dialogue unfolds after user asks for clarification and the system responds.

Task Progress: The extent to which the group moves toward a substantial, coherent, and evidence-based conclusion to the open-ended EDA task

To sum up..



RQ5: How do *users perceive the usefulness and interpretability* of these responses for the EDA task completion?

Perceived Usefulness: How smoothly the collaborative dialogue unfolds after user asks for clarification and the system responds.

Perceived Interpretability: The extent to which the group moves toward a substantial, coherent, and evidence-based conclusion to the open ended EDA task

Conclusion: Conversational Assistants for Collaborative Data Exploration

Data analysis need to be accessible for non-technical users and multi-user collaboration

CA that supported chart-centered collaborative exploration, and its design shaped how users moved through the task and interpreted what they saw.

Users asked clarification questions when they needed help understanding charts or deciding what to do next.

Clarification responses supported local interpretation and continuation of analysis within a broader chart-centered workflow.

